

Volume test — end-to-end on Cardano Preprod (2026-09-10)

Milestone 3 evidence: "UI flows function end-to-end in pre-prod" and the "List of transaction identifiers associated milestone test anchors" (the 180 TXIDs are in the appendix and, formally, in [transaction-ids.md](#)).

English translation of the working document `specs/REPORTE-2026-09-10-prueba-de-volumen.md`, as of 2026-09-21. D-NNN references point to decisions in `DECISIONS.md`.

Executive summary

3 new projects were created (Torre Volumen 1, Torre Volumen 2, Torre Volumen 3, 10 stages each from the default stage catalog) from the real front end, against production (`propnexus-web.onrender.com` / `propnexus-api.onrender.com` / Cardano Preprod), and **each of the 30 stages** was taken through the full tour of the stage state machine:

Pending → InProgress → Observed → InProgress → Completed

Every transition was triggered **by a real click in the browser** (Claude in Chrome), never by direct HTTP, with the demo accounts the flow requires (`developer@example.com` to upload evidence and "Resume stage"; `verifier@example.com` to "Observe" and "Certify"). It was the first time the four edges of the state machine were exercised **together**, on **10 stages per project**, repeated across **3 projects** — until then each edge had been tested once, on an isolated stage.

Result: all 30 stages ended Completed in the database. Of the **180 on-chain anchoring attempts** the test triggers (30 mints + 30 evidence anchors + 120 state transitions — 4 per stage), **176 had a real TXID recorded** (97.8%) on the first pass and **4 did not** (2.2%), concentrated in **2 of the 30 stages** (Torre Volumen 1, stages 3 and 4). The investigation of those 4 — confirmed against Cardano Preprod directly, not only against the database — found that **they are not the same failure**: in stage 3 the real transaction **was executed and confirmed on chain**, but the application never stored its TXID (leaving a live on-chain thread the database did not know about); in stage 4 the transaction **was never submitted**. See §Finding for the full forensic detail — it is by far the most important result of this test, because it exposed a real case of lost bookkeeping that no earlier test had exercised (two anchors triggered almost simultaneously).

Final state (two sessions, same calendar day as the finding): the root cause of the 4 failures was confirmed (a restart of `propnexus-api` after a failed health check on Render's free plan, unrelated to the code) and the gap that allowed them to be lost was closed in production (§Making it more robust — 3 code improvements, deployed). The 176 first-pass TXIDs were reconciled to `Confirmed` and **verified one by one against Preprod** through an independent source (Koios). The stage-3 bookkeeping was corrected without touching the chain (migration `0004_reconciliar_hilo_huerfano_etapa3.sql`), and the 3 on-chain transactions still missing (1 in stage 3, 2 in stage 4) were submitted in a following session — see §Closing the 3 pending transactions. **All 30 stages of the 3 projects are Completed, and the 180 on-chain transactions the test triggers have a real, Confirmed TXID.** The same session built, at the owner's

request, a self-healing layer so this failure mode no longer needs manual forensics — §Making it self-healing — Layers 0, 1 and 2. The service wallet's real balance was measured: **~99.8 ADA total cost** for the test (39.83 ADA in fees + 60 ADA locked permanently, D-057), slightly below the plan's estimate (~105 ADA).

Scope and goal

Repeat, at volume, what until 2026-09-08 had only been tested edge by edge:

Transition	Who	Mechanism (UI)
Pending → InProgress	developer, automatic	first evidence uploaded (/developer/project/:id/upload)
InProgress → Observed	certifier, exclusive	"Observe" button (/certifier/stage/:stageId)
Observed → InProgress	developer, manual	"Resume stage" button (/developer/progress, "Observed stages" section)
InProgress → Completed	certifier, exclusive	"Certify" button (/certifier/stage/:stageId)

Budget estimated in the plan: **~35 ADA per project** (≈15 in fees + 20 locked by the 10 stages × 2 ADA, D-057 — no burn, permanent).

Method

- **No HTTP shortcuts.** Each of the ~150 transition actions (30 stages × 4 edges + 30 evidence uploads) was executed with real browser clicks (mcp__claude-in-chrome) against https://propnexus-web.onrender.com, never with curl or direct API calls — as the owner explicitly asked ("everything by click in the browser").
- **Login per role**, with the production credentials: developer@example.com to create the 3 projects, upload the initial evidence and "Resume stage"; verifier@example.com (the certifier role) to "Observe" and "Certify". verifier@example.com was added as a member (membershipRole: "verifier") of the 3 new projects via POST /projects/:id/members (admin-only), because the certifier's queue only lists stages of projects where the certifier is a member — without that, the 30 new stages were invisible to the certifier. (Since 2026-09-21 this is done from the admin screen: the admin invites the certifier and the certifier accepts — decision D-095.)
- **Real evidence:** 10 PDFs generated with pandoc (acta-1.pdf ... acta-10.pdf), reused across the 3 projects, uploaded through /developer/project/:id/upload.
- **"Don't trust the spinner" verification:** after each transition click (especially "Observe"/"Certify"/"Resume stage"), the state was confirmed by reading the production database directly (turso db shell propnexus "SELECT state FROM Stage WHERE id='...' ", polling for up to ~2 min) before moving to the next stage. It was adopted because clicks fired without this wait gave false positives: clicks that "looked" successful in the UI while the request was interrupted by the following navigation.
- **The progress screen has no per-stage URL** for "Resume stage" — it is a list that reorders itself as each action completes, so phase 3 was done by always clicking the first button of the "Observed stages" list and confirming against the database which stageId it corresponded to at each step, with a full page reload between clicks.

Timeline

The five phases of the test, with real start/end times taken from `OnChainEvent.createdAt` in the production database (UTC):

Phase	What	Start	End	Duration
0	Creating the 3 projects (30 mints, <code>STAGE_CREATED</code>)	15:24:30	15:43:41	19m 10s
1	Initial evidence upload + automatic <code>Pending → InProgress</code> (30×)	15:45:45	16:21:18	35m 33s
2	"Observe" — <code>InProgress → Observed</code> (30×, certifier)	16:24:14	16:49:30	25m 16s
3	"Resume stage" — <code>Observed → InProgress</code> (30×, developer)	16:50:56	17:05:55	14m 58s
4	"Certify" — <code>InProgress → Completed</code> (30×, certifier)	17:07:01	17:18:27	11m 26s

Total duration: 1h 53m 57s, end to end, for 180 real anchoring attempts against Preprod (slower at the start of each phase because of Render free-tier cold starts).

Aggregate results (first pass)

By on-chain event type

Event	Attempted	With real TXID	Without TXID
<code>STAGE_CREATED</code> (mint, 1 per stage)	30	30	0
<code>EVIDENCE_ANCHOR</code> (initial evidence, 1 per stage)	30	30	0
<code>STAGE_TRANSITION</code> (4 edges × 30 stages)	120	116	4
Total	180	176 (97.8%)	4 (2.2%)

By state-machine transition

Transition	Attempts	With TXID
<code>Pending → InProgress</code> (automatic)	30	30
<code>InProgress → Observed</code> ("Observe")	30	30
<code>Observed → InProgress</code> ("Resume stage")	30	28
<code>InProgress → Completed</code> ("Certify")	30	28

By project

Project	Completed stages (declared, database)	Real TXIDs obtained
Torre Volumen 1	10/10	56/60
Torre Volumen 2	10/10	60/60
Torre Volumen 3	10/10	60/60

All 30 stages ended declared `Completed` in the database, including the 2 with failed anchoring — consistent with D-059 (the declaration is written even if anchoring fails) and with the permission matrix: no transition was rejected for authorization, all 30 went through the correct role (developer for the automatic

and manual edges that belong to it, certifier exclusively for Observe/Certify). The 4 missing transactions were closed afterwards — see below.

A note on Pending vs Confirmed

By design (D-077, reconciliation on read), `OnChainEvent.status` only moves from Pending to Confirmed when a route that reads it triggers reconciliation — not automatically. Right after the test, most events showed Pending with a real TXID: that is **not the same as "not confirmed on Preprod"**, it means "written and with a TXID, but nobody has opened the screen that reconciles that particular event". A reconciliation pass afterwards left all of them Confirmed (see §What was left after this report, item 3).

Finding: 2 of 30 stages with failed anchoring (Torre Volumen 1, stages 3 and 4) — root cause confirmed

Torre Volumen 1 · Stage 3 ("Movimiento de suelos y excavación", earthworks and excavation) and Stage 4 ("Cimentación", foundations) are the only two of the 30 where real anchoring did not complete — in both, at the same point of the sequence (event 4, "Resume stage", `Observed` → `InProgress`) and with the same knock-on effect on event 5 ("Certify", Failed). **The root cause is confirmed**, by crossing independent sources: Cardano Preprod directly (Koios, `preprod.koios.rest`), the `propnexus-api` logs on Render (`render logs`) and the deploy history (`render deploys list`).

The cause: `propnexus-api` went down in the middle of phase 3 — free plan, no clean shutdown

```
16:52:06 Resume stage 1 and 2 of Torre Volumen 1 – confirm on chain, normal
16:54:13 click "Resume stage" → stage 3 (event recorded, Stage.state becomes InProgress)
16:54:45 click "Resume stage" → stage 4 (event recorded, Stage.state becomes InProgress)
16:54:56 the stage-3 transaction CONFIRMS on Preprod (block 5161038) – the process that
        triggered it is no longer there to find out
16:55:15 Render starts a NEW propnexus-api instance (logs: "Running '...migrate.js...
        server.js'", with no "[SIGTERM] shutting down" before it – unlike the clean
        shutdown seen later, at 17:40:03)
```

`render deploys list` confirms **there was no deploy at that moment** (the last one finished at 15:16:38 UTC, almost 40 minutes earlier) — the 16:55:15 restart was not a deliberate redeploy. And `render service s` confirms the service's plan is `free` (512 MB of RAM, a single instance). **Confirmed by a fourth source, the most authoritative — Render itself:** the owner received Render's standard email at 16:55 UTC, "we detected a server failure, HTTP check failed (timed out after 5 seconds)", matching the restart in the logs to the second. The platform's declared cause is that `/health` **did not respond within the 5 seconds** Render tolerates before killing and restarting the instance. The absence of a clean shutdown line is consistent with this: if the process was too busy or blocked to answer a trivial health check, it also had no room to shut down in order.

The exact blocking mechanism could not be pinned down. Render's metrics API was queried for `propnexus-api` CPU and memory in the 16:40–17:00 UTC window:

- **CPU:** never above ~8% of one core in the whole window, and in the exact minute of the failure (16:53–16:54 UTC) at its **lowest** point of the test (<1%). This rules out an event-loop block by heavy

synchronous computation (building/signing transactions, evaluating the Plutus script) at that moment.

- **Memory:** rising steadily throughout the test — from 300 MB (16:40 UTC) to 331 MB, its peak, at 16:54 UTC — against the free plan's 512 MB limit (64%). It did not reach the limit, so it was not a hard out-of-memory kill. The sustained growth — which climbs the same way after the restart, starting from ~131 MB — does suggest a memory leak under sustained anchoring load, independent of what triggered this particular restart.
- **Per-request HTTP latency** (which would have shown whether an I/O call — Blockfrost or Turso — hung and blocked the health check) **is not available on the Free/Hobby plan**. Without it, a hung external call cannot be told apart from a transient network problem on Render's side.

In short: the data rules out the two most obvious causes (high CPU, memory at the limit) without closing the exact cause — the current plan does not provide the telemetry for that. It does not change the conclusion or the improvements below, which close the real problem (losing the receipt of a successful anchor when the process dies midway) whatever the specific trigger was.

Exact consequence for each stage, depending on how far the process got before dying:

- **Stage 3** — the old process managed to **build and submit** the "Resume stage" transaction to Blockfrost before going down. Blockfrost accepted it and Preprod confirmed it seconds after the new instance was already starting — but the process that had built it no longer existed to run the `UPDATE OnChainEvent SET txid = ...` that follows the `await` on the chain. Event 4 was left with `txid` and `outputRef` `NULL`, while the real transaction lives, confirmed, at `9a57f563e7d5627f22d9a062d26049d48789c91014d8b676298c0eb7c71f9e68#0`. **The anchor did not fail: it succeeded, and the process that had to record it did not get to.**
- **Stage 4** — the process died before its request got that far: its known UTxO (`270cfea4...#0`) remained unspent on Preprod — no transaction was ever submitted. This one is a simple failure.

The new instance's logs prove the knock-on effect on "Certify" (17:07–17:08 UTC, ~12 minutes later, already in the new process):

```
2026-09-10 17:07:48 [anchor] el anclaje falló {
  eventId: 'd8rlw6iu9kmp8bma2nw38ycd',
  error: AnchorRejectedError: No existe el UTxO fc37cac7dd081198629a464d038e69f8130ba123295c
cf2043775426cce7ea65#0
  code: 'UNKNOWN_THREAD'
}
```

That is **stage 3** ("el anclaje falló" = "the anchor failed"; "No existe el UTxO" = "the UTxO does not exist"): it tried to spend the old `outputRef` (`fc37cac7...#0`) because the database had no newer `outputRef` to return (event 4 never wrote it) — and that UTxO had already been spent by the orphan transaction.

```
2026-09-10 17:08:11 [anchor] el anclaje falló {
  eventId: 'v8w15f2q7wxvjc8inod5tn4d',
  error: EvaluatorError: "failed script execution Spend[0] the validator crashed / exited pr
ematurely"
}
```

That is **stage 4**: here it did find a UTxO (the only one, `270cfea4...#0`, still `observed` on-chain), but built the transaction assuming the previous datum was `InProgress` — because that is what `Stage.state` said in

the database (D-059 had already written the declaration) — and that rebuilt datum **did not match** the real datum on-chain. The Aiken validator rejected the transaction because the declared previous state was not the real previous state. The validator worked exactly as designed — the problem was that the database and the chain no longer agreed on the previous state.

Real impact

Because the 4 phases of the test ran over all 30 stages equally, stage 4's `Stage.state` also ended `Completed` — terminal in the state machine (D-020) — so no UI action could repair either stage. Both required the same thing: writing the real `outputRef` into `OnChainEvent` and building by hand the missing anchoring transaction. That was done in the same day's second session (below).

What was left after this report — all done

1. **Repair the orphan threads of stages 3 and 4 — done.** The UI could not do it (both stages were already `Completed`), so it was resolved against the anchoring port directly, with the real production credentials verified first (deriving the wallet address from the private key, offline, and comparing it with the one that produced the 176 first-pass transactions). **Stage 3:** bookkeeping corrected without touching the chain (migration `0004_reconciliar_hilo_huerfano_etapa3.sql`) and the missing "Certify" transaction submitted. **Stage 4:** the 2 transactions that had never been attempted, submitted. Detail and the 3 TXIDs in §Closing the 3 pending transactions.
2. **Confirm the cause with server logs — done.** See §The cause above.
3. **Run a reconciliation pass — done,** on 2026-09-10, after deploying the fix. `POST /evidence/reconcile` run by hand (admin) against production: **72 events** moved from `Pending` to `Confirmed` — together with those already confirmed, **the 176 real transactions of the test are Confirmed in the database.** The response also returned `sospechosos` ("suspicious", improvement #2 below): besides stages 3 and 4, a **third, pre-existing case** appeared, from before this test (2026-09-08, a different project, not one of the Torre Volumen ones), later closed as well (see §Closing the 3 pending transactions).
4. **Verify the appendix against the chain — done. The 176 TXIDs were verified one by one against Preprod via Koios** (`preprod.koios.rest/api/v1/tx_info`, in batches of 40) — **176 of 176 really exist on chain**, zero missing. It is a source independent of the production database: not "the app says these TXIDs exist", but "Cardano Preprod says these TXIDs exist".
5. **Measure the service wallet's real balance — done.** Current balance of `addr_test1vp3vy56p6lrgghntg8ytydnuugnqh7ctkyxn3rm35g4q2ggtqvncw`: **9,870.97 ADA**. No "before" balance was captured, so the most honest measure is the one rebuilt from the transactions themselves: summing the real fee of the test's 176 transactions gives **39.83 ADA in fees**, plus **60 ADA locked permanently** (30 live threads × 2 ADA, D-057, no burn) — **≈99.8 ADA total real cost, ≈33.3 ADA per project**, against the plan's estimate of ~35 ADA/project (≈105 ADA for the 3). **It came in slightly under budget.**

Making it more robust and transparent — ✓ implemented on 2026-09-10

The confirmed root cause is an infrastructure event (Render killing the process over an unanswered health check) — but **the reason for a restart can change at any time** (a deploy, a real out-of-memory, any other crash, a manual `kill -9`), and the underlying problem was that **the code did not survive the process dying between "the transaction went to the chain" and "its receipt was stored"**. That window always

exists, whatever causes it. Three improvements, all implemented and verified the same day (full `pnpm verify` green):

1. **✓ Close the loss window, not just detect it.** Anchoring no longer waits for confirmation before storing `txid/outputRef` — the `UPDATE` with the receipt runs as soon as the chain call returns, and confirming became a best-effort second step: if it fails or the process dies there, the event stays `Pending` with a real TXID (never `Failed`), and reconciliation confirms it later. The same fix was applied to the two other call sites with the same logic. New test in `test/stage-transitions.test.ts` that reproduces stage 3's exact scenario (confirmation throws; the TXID is still written with `status: "Pending"`, never `"Failed"`).
2. **✓ State-based detection, adapted to a project with no cron or `setInterval` inside the API.** `hiloSospechosos()` ("suspicious threads") is a query, not a job, that finds every `STAGE_TRANSITION` without a `txid` whose stage already has a newer event — exactly the shape of the stage 3 and 4 finding. It was added to the response of `POST /evidence/reconcile`. Three new tests in `test/reconcile.test.ts`.
3. **✓ Make a failed or lost event visible without having to look for it.** The three catch blocks for failed broadcasts now call `Sentry.captureException` as well as logging — before, it only stayed in Render's log, with the short retention this same session ran into while investigating.

What was deliberately not proposed: upgrading Render from Free to a paid plan to avoid the health-check kill itself. It is the *trigger* of this particular case, not the underlying cause — the same problem can happen for any other reason, and with fix #1 it no longer matters which — and it is a cost decision, not a code one.

The code changes were reviewed, committed and pushed the same day (commit `56556c6`, deploy confirmed `Live` on Render).

Repairing stage 3's orphan thread — bookkeeping corrected

Correcting stage 3's bookkeeping (so the database points to the real UTxO) anchors nothing new — it corrects data that already existed on chain, without touching the chain.

Checked before touching the database: that the local credentials were the real ones. The wallet was verified **by deriving the public address from the private key, offline**, and comparing it with the address that produced the test's 176 real transactions: they match.

Dry run first: the correction was wrapped in `BEGIN; UPDATE ...; SELECT ...; ROLLBACK;` in a single `turso db shell` invocation, to see the row exactly as it would end up without committing. Once the result and the rollback were confirmed, the real `UPDATE` was run.

The change is versioned, not a loose `UPDATE`: `apps/api/migrations/0004_reconciliar_hilo_huerfano_etapa3.sql`, with an idempotency guard (`AND txid IS NULL`) so it does nothing on any other database or when re-applied.

```
-- apps/api/migrations/0004_reconciliar_hilo_huerfano_etapa3.sql
UPDATE `OnChainEvent`
SET `txid` = '9a57f563e7d5627f22d9a062d26049d48789c91014d8b676298c0eb7c71f9e68',
    `outputRef` = '9a57f563e7d5627f22d9a062d26049d48789c91014d8b676298c0eb7c71f9e68#0',
    `network` = 'Preprod',
    `status` = 'Confirmed',
    `blockTimestamp` = 1789059296000,
    `updatedAt` = unixepoch() * 1000
WHERE `id` = 'amc9u0gyovc9tlf5z9ceh4db' AND `txid` IS NULL;
```

Result, verified against production: `POST /evidence/reconcile` no longer lists stage 3 among the suspicious threads.

Closing the 3 pending transactions — all 180 confirmed

The 3 missing transactions were submitted against Preprod with the same verified credentials:

Stage	Transition	TXID
Torre Volumen 1 · Stage 3	InProgress → Completed ("Certify")	8753cd70211c4f8182da14d25375f81218d8a0568975e4ad0d5e76557ab148a4
Torre Volumen 1 · Stage 4	Observed → InProgress ("Resume stage")	cf23d689952846413a4efedea45c4d148e8a9284796ae47e2010c05a84e10f4a
Torre Volumen 1 · Stage 4	InProgress → Completed ("Certify")	75b9dcfd267848cefcc7a150f84b816524769736dfca7694012f53380d341ba8

How they were sent: a one-off script (the precursor of the Layer 2 tool below) that built each datum and called the anchoring port against the real adapter. It did not touch the database — it printed the `UPDATE OnChainEvent` for each step, to be run separately after review.

A stumble along the way, without repeating the bug this session came to close. The first attempt (stage 3 certify) crashed *after* the transaction had already gone out — a `JSON.stringify` over an internal adapter object carrying a `BigInt` —, exactly the window the robustness fix had closed in production code. It was resolved with the same forensic discipline: the real transaction was looked up on Koios (the wallet's `address_txs`, ordered by `block_time`), its inputs/outputs verified and its resulting datum decoded against what was expected before accepting the TXID. The script was fixed and the two remaining steps ran without problems.

Verified against production after the three: `POST /evidence/reconcile` returns no suspicious thread for Torre Volumen. **All 30 stages of the 3 projects are Completed in the database, and the 180 on-chain transactions the test triggers (30 mints + 30 evidence anchors + 120 transitions) have a real, confirmed TXID.**

The pre-existing 2026-09-08 case, closed the same day Layer 0 surfaced it. That event (on the older `torre-a` project, InProgress → Observed, Failed, no `txid`) belonged to a stage seeded directly in the database before the stage template existed, which **never had a real on-chain thread** — so it could never be repaired and would have reappeared as suspicious on every run. It was removed from production (migration `0005_borrar_intento_sin_hilo_estructura_torre_a.sql`, same dry-run and idempotency-guard pattern). `POST /evidence/reconcile` afterwards: no suspicious threads.

Making it self-healing — Layers 0, 1 and 2 (✓ implemented on 2026-09-10)

The finding showed that **detecting** a suspicious thread is not enough if nobody runs the reconciliation — and that, even then, the only possible repair was the manual forensics above, repeated by hand for each case. Three layers, each solving the part the previous one leaves out:

Layer 0 — someone finds out, without having to remember to look

`.github/workflows/reconcile.yml`: a GitHub Actions cron (now daily at 06:00 UTC, plus `workflow_dispatch` to run it by hand) that logs in as admin and calls `POST /evidence/reconcile` against production. If the response contains a suspicious thread that Layer 1 could not repair on its own, the job **fails on purpose**: a red run in Actions triggers GitHub's default notification, with no new alerting channel for a problem that is rare today.

Why GitHub Actions and not a cron inside the API: decisions D-003 · D-040 · D-077 forbid it — an internal timer stops counting as soon as Render puts the service to sleep after 15 minutes of inactivity (free plan, no workers). The cron fires from **outside** the process.

Layer 1 — repair the bookkeeping automatically, signing nothing

Stage 3's case — a transaction that **did** go out and confirm, with the database simply unaware of it — can be repaired with no risk: nothing new needs to be built or signed, only the real UTxO found and the lost `txid/outputRef` filled in. The anchoring port gained `findLiveThread(stageRef)`, which locates a stage's live thread by the asset of its thread token (D-058), implemented in the real, simulated and disabled adapters. `repararHilosSospechosos()` ("repair suspicious threads") uses it: for each suspicious thread it finds the live one and, **only if the state in the datum it finds matches the state the event already declared**, fills in `txid/outputRef` — leaving the status `Pending` for regular reconciliation to confirm. If the datum does not match — stage 4's case, where the transaction never went out — nothing is touched: that repair stays manual by design.

`POST /evidence/reconcile` now runs `repair`, `reconcile` and `list-suspicious` in that order. Covered by tests in the simulated adapter, the real adapter (against Lucid's emulator, with the real validator) and `reconcile.test.ts`.

Layer 2 — the tool for when signing is actually needed

What Layer 1 never does on purpose: submit the missing transaction when the real thread is **not** in the required state (stage 4's case). That still signs with the service wallet key and spends real ADA, so it is a manual tool, `apps/api/scripts/repair-thread.ts`, built at the owner's explicit request that there be "no risk of spending ADA":

```
pnpm --filter @plataforma/api exec tsx scripts/repair-thread.ts \
  --stage <stageId> --to <InProgress|Observed|Completed> \
  [--evidence-root <64 hex chars>] [--completed-at <epoch ms>] [--confirm]
```

It reads the stage from the database, finds its live thread with `findLiveThread` (Layer 1), validates the transition against the state machine and builds the next datum. **The risk removed is not that the tool can sign — it is that it signs by accident:**

1. **Without `--confirm`, it is a dry run:** it prints the live thread found and the datum it would build, and stops there.
2. **It is not a route or a job.** It lives in `scripts/` and only runs by hand. No production code imports it.
3. **It does not touch the database.** It prints instructions for the later reconciliation instead of writing `OnChainEvent` directly — that write remains a human review.

Appendix — TXIDs per stage (180/180, all 180 confirmed on chain)

The 176 first-pass TXIDs were verified individually against Cardano Preprod via Koios, not only against what the database says. The 4 missing ones from Torre Volumen 1 / stages 3 and 4 were closed afterwards (one was pure bookkeeping — the transaction was already on chain — and the other 3 were submitted), verified against the production database: the 6 rows of each of the two stages are `Confirmed`. Stage names are kept in Spanish, as they appear in the product (default locale `es-AR`).

Grouped by project and stage, in state-machine order.

Torre Volumen 1

Stage 1 — Adquisición del terreno

Transition	Database status	TXID
Mint (declaration)	Confirmed	3527dae63d58996878f59dbc83f91903173e8943b2d5e3e9e4da12feccd0cde96
Evidence anchor	Confirmed	bce9a2ac69130202ffdc6cd580b6f63fb102093d92eec2e095883172d0722485
Automatic Pending→InProgress	Confirmed	e82bc35a335cbfe869e05def53c29322e95496a84043794e5a72ec613f198ae1
Observe (InProgress→Observed)	Confirmed	a27f329fe7aad4456425ca59489ffd26071a06d771924f66f4c880ac14df2b6c
Resume (Observed→InProgress)	Confirmed	e7f9c7b7d88bf9d84c5931a35f40742c4a1a6fb69a5081aa797a7be91ee23e95
Certify (InProgress→Completed)	Confirmed	3a1f95cf82ac4e12c267e3645439e3e8013c4548fe754d4ac203e5b1044e68eb

Stage 2 — Proyecto ejecutivo

Transition	Database status	TXID
Mint (declaration)	Confirmed	ca6793e071de3b19dbc25136e6868ce3adf5a46c0fe4d8b77e3389c282cc7573
Evidence anchor	Confirmed	b377f2b157b47f9a98d02c418f7423dc08613dd62bda6b03cfd1b26cb37e9ece
Automatic Pending→InProgress	Confirmed	7412e42b9403a4e4051d2adcb7782b76e61ec117cee45d01f1621bdf264b8b47
Observe (InProgress→Observed)	Confirmed	5bef86563f8cfb543aa448fe91bbd4ed565ac2d21678adf88f33e8364ea02289

Transition	Database status	TXID
Resume (Observed→InProgress)	Confirmed	ad8b6633882a6f3cbc3e367f46211be40c6101fc4f862934d5902a72b828ed53
Certify (InProgress→Completed)	Confirmed	cf0e14652205163b33309519227bf341a490e78b80ddac0a06503fdf40fd7c27

Stage 3 — Movimiento de suelos y excavación

Transition	Database status	TXID
Mint (declaration)	Confirmed	61067ba2fe920bfbb350559ba7007a7beed7dd9790713878df3fd0ba3b3b6240
Evidence anchor	Confirmed	0132d1497f2e63247bd78d13728c46c31748114bb3de7e838535c9e428eafb2c
Automatic Pending→InProgress	Confirmed	49a6be960d61ba55ba4d16de9b766676d4da155581dcbccace52621e2fe4dbf6
Observe (InProgress→Observed)	Confirmed	fc37cac7dd081198629a464d038e69f8130ba123295ccf2043775426cce7ea65
Resume (Observed→InProgress)	Confirmed	9a57f563e7d5627f22d9a062d26049d48789c91014d8b676298c0eb7c71f9e68
Certify (InProgress→Completed)	Confirmed	8753cd70211c4f8182da14d25375f81218d8a0568975e4ad0d5e76557ab148a4

Stage 4 — Cimentación

Transition	Database status	TXID
Mint (declaration)	Confirmed	ad0fe12703dd89470f4ad7b9ee69b9e681dc0434f14edc685a673c2f191609e4
Evidence anchor	Confirmed	46bf2b8c64b10eb26a99005d888339c423ae692eee5d3b02fddfe9d5e40737e7
Automatic Pending→InProgress	Confirmed	1972aa5e68d4fe800a6542f16f956b7352d8f7bd62752dedd3c5b725f89a12a4
Observe (InProgress→Observed)	Confirmed	270cfea4fabf232573e94b69324b69054f41f344fbaeca6b95dfe5d16595dd06
Resume (Observed→InProgress)	Confirmed	cf23d689952846413a4efedea45c4d148e8a9284796ae47e2010c05a84e10f4a
Certify (InProgress→Completed)	Confirmed	75b9dcfd267848cefcc7a150f84b816524769736dfca7694012f53380d341ba8

Stage 5 — Estructura planta baja

Transition	Database status	TXID
Mint (declaration)	Confirmed	3319e8e26babbb02ecabb14214c956abf6b7e79ccbb9de410817362562e1e25d

Transition	Database status	TXID
Evidence anchor	Confirmed	113e729ec564b8f3c748145bd0645a2b56fbfc5465420d1e20f116f843addfc
Automatic Pending→InProgress	Confirmed	5736ced65bb251ca350fb1d9e31cafffac192906e3dcd94d1675914127162576
Observe (InProgress→Observed)	Confirmed	0fb0feb61c995364d63b023c94a37e0372ca0d1d2a11db0569dc90937236b2a4
Resume (Observed→InProgress)	Confirmed	cbf60e0a3ceb830ff8fcc80a6cabcb876887c56118b2c4f3e0d8dc04e51edca3
Certify (InProgress→Completed)	Confirmed	636260b0b67c0fc9f56a5baef91864c82221bf2dde0f653dba119ca0d3ba56f6

Stage 6 — Estructura niveles superiores

Transition	Database status	TXID
Mint (declaration)	Confirmed	f24667903957ee5f1e861bfff00e6d754a21e1b7ae8c2c8cd18862fa0d745f482
Evidence anchor	Confirmed	b6e9e7d651d15a69d7757a039ec2dc56ba4b82352a300ae545aa09b6deb4b6ad
Automatic Pending→InProgress	Confirmed	fddf96d86bab1f01b5db73c000b9e571ee9fdc88841873af5056458e073fa4de
Observe (InProgress→Observed)	Confirmed	0657eebfa52e3775595c9f1b2ad5ff5bb989c1decf359355674325b4c7f238a4
Resume (Observed→InProgress)	Confirmed	d5e8f54c298f8fb0f42a16685090526af1466a55e5a5ed733e065c47cecc1556
Certify (InProgress→Completed)	Confirmed	c3c0bd1925eaa0554bee0daf3e0ff05e84787ee377ba22d909d20f4e437a82fc

Stage 7 — Cerramientos y mampostería

Transition	Database status	TXID
Mint (declaration)	Confirmed	3759177e0e18f520b02454d3df9bb9066aa2f58cb37a4ca4a61615171c6c3ed1
Evidence anchor	Confirmed	4aa94e5207574700a1adc214ac9551c947fc76151e248eea02b4d4e57263411b
Automatic Pending→InProgress	Confirmed	17b8ba858673a85930a20c98f25a4cf9ab0ac11715b6924b3e22b9c4e82c015f
Observe (InProgress→Observed)	Confirmed	fac73f94fad34601f9cbdc2ad8b0b4ad8bfff371248430c762dcde6ff2f0414c2
Resume (Observed→InProgress)	Confirmed	06f51b9cd7f773b3b68dcb95c1ff1039e53458d63703b3bb304bfee44610c689
Certify (InProgress→Completed)	Confirmed	5b080c1fffd8988c6f60603334bbc9447ce31fc3944c873b1fba5654205e81e70

Stage 8 — Instalaciones

Transition	Database status	TXID
Mint (declaration)	Confirmed	0b71997e680748e92cf06e09db8d65fe17cc625f2e3af046d5b0a869aff47189
Evidence anchor	Confirmed	f73af36678a0f54ee50ce54dec25ab9899c2a018c14c4dc6659e5f0905a13ab8
Automatic Pending→InProgress	Confirmed	fd78664b3b6599ff99097f1fd2e7bd99c70cf9afbecd4fe2d1bbc071a8b853fd
Observe (InProgress→Observed)	Confirmed	cb46248e9fed481262dda1d514e05b26db8dbf337e747dc5203c397794eabdbb
Resume (Observed→InProgress)	Confirmed	dcef912606d674522fb6a581e14d48a457683b1ea77898646486f1138a6615b6
Certify (InProgress→Completed)	Confirmed	ead1aa198a77e19bb2d73bc3529203e08afe8b50aae334edf66b766253f5dfca

Stage 9 — Terminaciones

Transition	Database status	TXID
Mint (declaration)	Confirmed	085ebd2ec2b589ac2baf8dbae106118c74e27d24a02eb9811078f0b5cfe2c6fb
Evidence anchor	Confirmed	6c6c1e208285c9395a099d0f5502e3d9ab5083adbc8629068decf9241254012d
Automatic Pending→InProgress	Confirmed	c837dd192222c49ee971a9d8188195dd35d4216422d8b2fb12cc3a9d0bdb37d0
Observe (InProgress→Observed)	Confirmed	531cb045476bce32010fb54e588e48566b871aebfd9b1a05f05927a2a3eed82b
Resume (Observed→InProgress)	Confirmed	2cbb0bc5b1abac1cee1120499fc4c28fb78749f8ccc2c3a8e39e1784a7f9aeba
Certify (InProgress→Completed)	Confirmed	9ebcf753e8c88ba6b4c3aed1de2b41bd968aa1d74a66e325fb83aaf68ae3bc5b

Stage 10 — Final de obra y subdivisión

Transition	Database status	TXID
Mint (declaration)	Confirmed	fe3d21b358bf931e5f78dbc559f66cc2b2b341a9c5a62dc95c504dd6ef0af2b2
Evidence anchor	Confirmed	36fce09bd1a041b41fbb04fa573b1f4f253e95e0cd62d0bc2cf9dd010e346dad
Automatic Pending→InProgress	Confirmed	831a17ab5b34365437787729cdb2750a90a033af858ff839738c6e9a665fd9e7
Observe (InProgress→Observed)	Confirmed	2e5f84b76f0a9f950cad20018d1aac526981044e1c135cd2bc9aa9bc15873609

Transition	Database status	TXID
Resume (Observed→InProgress)	Confirmed	aa1abfd70a388fcb26eb2e3ffab3d12e6450db4e8c866bd7d286a85ea333714e
Certify (InProgress→Completed)	Confirmed	ab9794a5010eb0d6f672cd0e9bc45f1499a8e38f904e3ca080625c7714b95ba7

Torre Volumen 2

Stage 1 — Adquisición del terreno

Transition	Database status	TXID
Mint (declaration)	Confirmed	d253922c485274b6c4cc79069564fc25ffc0fd9380ef5944d6963462437cf0af
Evidence anchor	Confirmed	f4df818e30b64f7c74fbcfd54308a83a5d0648fb2018701bbe33b43d6fdaca33
Automatic Pending→InProgress	Confirmed	e6df4aba0a3fb38179cd69b9eb817e3fdd06dbf0aa6ca67174cf065ac835e72f
Observe (InProgress→Observed)	Confirmed	5bf3fc45f50b9c91be9ebe3edde5565b6938dcfffd7d76065f3d4832dfc5c4125
Resume (Observed→InProgress)	Confirmed	c9227010477854f49378182140753ea3a5a8dcd67ff98d480bd4d5b7864bc166
Certify (InProgress→Completed)	Confirmed	6b50c5ca5ede70b2963b6c395e3d4b51ead85f6588b2ad07a91394597e9bcab6

Stage 2 — Proyecto ejecutivo

Transition	Database status	TXID
Mint (declaration)	Confirmed	9b14ec76069b288397955f48fd40f5f2ae532c456be4c2e0d0b6860dcd39d6a2
Evidence anchor	Confirmed	d4e4734b359498ae890865f40b3df84d7de5b22be792f547b51f3886a7c06d9d
Automatic Pending→InProgress	Confirmed	0d9034de19a21a23accef7e0054147ea8569a8f33610681e40de1defe110e30a
Observe (InProgress→Observed)	Confirmed	e1d0528b8ec48ed2f601d11ffda0250143153c66bbc2bf424a293114e13fc92c
Resume (Observed→InProgress)	Confirmed	c2768c1e712d7e6f42484a56ceb2de5bdd1a3684daa1fcdff00dcea6958dd798e
Certify (InProgress→Completed)	Confirmed	2f754af5c1b445d300ed81bc8a5ece986516518bdacbd17e072baf4cb19258a3

Stage 3 — Movimiento de suelos y excavación

Transition	Database status	TXID
Mint (declaration)	Confirmed	d5fced405b5b3f256c0a33b19cce9284ef76d321086eb41315ebb8cb329e2aa2
Evidence anchor	Confirmed	b36aa79c33c85fbef36c25f270c8f7ac28521d75f3579b163ef01c8dd07b1aa2
Automatic Pending→InProgress	Confirmed	ac7bd08fd8bee1bda6ddce08f89a6f0ff1dcfe98e1b14196777bb8f9df9d792c
Observe (InProgress→Observed)	Confirmed	ea8c78ef3244e8551af2e42db2a37fb40a4d14b091d6aa4797c0baf12385e03a
Resume (Observed→InProgress)	Confirmed	d451bbbce026434ecb2a76b38dd516ad63b24835f24a9522631bd270b00c78b5
Certify (InProgress→Completed)	Confirmed	a5c90d6cef0dd15f1553ad3a40437e7e008e32e99963883d4e9d551358724f0f

Stage 4 — Cimentación

Transition	Database status	TXID
Mint (declaration)	Confirmed	5fa9493e2208ed94696354d7ca7374acb97a2683bb496db46d91a96b39f422fc
Evidence anchor	Confirmed	5cd74d393fd69e74d84eb156d66e4c40f7b489a9f003399a8d68a248918cc0c0
Automatic Pending→InProgress	Confirmed	286586df2416fa21d83dd4aa17b40ab6dbb900ddd835adf6a3beada2ee9bebaaf
Observe (InProgress→Observed)	Confirmed	80c693dfcb59abc77128c7a276750e27d1c0e80f89a6103f7178f02d58d5c543
Resume (Observed→InProgress)	Confirmed	255bfd5c4f577ec14031e920d4520e2f9656fb822fa07402f68343199d987f47
Certify (InProgress→Completed)	Confirmed	0238a9ae5ae6bb5ac96595ca9a58eca215d52b2f0106bc155096a163c12b2d25

Stage 5 — Estructura planta baja

Transition	Database status	TXID
Mint (declaration)	Confirmed	d2a43fe211837f40b1c273bd0196fbc5dd67b329c7e0cee45883e210efff398
Evidence anchor	Confirmed	c9f7e5817a409c2850ed8b082830de1542aabf2c382d46e9dfef48e7cd4f49a7
Automatic Pending→InProgress	Confirmed	33f7deb053da97125c4ba61ab4b70af8ba00c3a9ecb9f556012275ddade0463a
Observe (InProgress→Observed)	Confirmed	ae376b86bd727974574c77a007bd07e0f69b24963bbc2ae778b2a2975efc278f
Resume (Observed→InProgress)	Confirmed	0069507b3c1aba88ad4852cf56265c2bf729b7857f35458d4d495062533c9455

Transition	Database status	TXID
Certify (InProgress→Completed)	Confirmed	3031b0198fce05fe9b25efa5e3b28138ab06bc1baf00a00c86599ca d527102bf

Stage 6 — Estructura niveles superiores

Transition	Database status	TXID
Mint (declaration)	Confirmed	de5e195779b814a20b9a37938ab245f0c0e9b6b599683c9d564f6a6 9439ff235
Evidence anchor	Confirmed	40fd381daecbc1239c73afd07ae0d8907c39a6e99d77dc063158b60 8f1ac50c7
Automatic Pending→InProgress	Confirmed	c71d3593cacbd643498c26602fd557b4527c692f350c5ff35a76e95 af04e3f0c
Observe (InProgress→Observed)	Confirmed	9d7696c5523f83fb3ebe2c8df7ba75c35eaaa92d53cc92c21468c68 382b41dc9
Resume (Observed→InProgress)	Confirmed	a9cf646aca7a1b4d0fe16b31dbf2b8ed55f266689ea443d0b7c52ef b31f70f6e
Certify (InProgress→Completed)	Confirmed	cad2aa46265feefc9eadc0324fe139d94e3192ebd89898a13cfce4a 453e77c85

Stage 7 — Cerramientos y mampostería

Transition	Database status	TXID
Mint (declaration)	Confirmed	633a5f5980c845f408268a536d310083363ae56dd0b27997e8c3a84 57e594952
Evidence anchor	Confirmed	158155667d6fb5fdc137407f3b998c75a8a14f0030c12f5457be508 4c1614385
Automatic Pending→InProgress	Confirmed	38b669b2f33c33d958b8f6f4ecafe2247045b05ada288aa8d1a2078 0bfbf84aa
Observe (InProgress→Observed)	Confirmed	19ebb84a3808df8417577c7dcef4292d246dfed0a4c363d1596526b 1ed214c2d
Resume (Observed→InProgress)	Confirmed	102cc832b73b5aa3c95f7adc339d4c170a2ece3a7a8076df7090795 909aa4814
Certify (InProgress→Completed)	Confirmed	315dbbc13bae27a06ef2af9025a6f637bbe6f7f53ede69b41f2c223 c0b775d23

Stage 8 — Instalaciones

Transition	Database status	TXID
Mint (declaration)	Confirmed	735f7fab3ae469793fa1a4028f58c107f81849abd84ae60911d1f85 2222d1352
Evidence anchor	Confirmed	4a37d7aaf2b6a51ea236e38bc7316a33d6aa944fc3f9b76497dfa93 244cfeb4f

Transition	Database status	TXID
Automatic Pending→InProgress	Confirmed	5065f1bfc75da300f487d69fcc434df504010aa3127aec150291d1672f4aecab
Observe (InProgress→Observed)	Confirmed	7cc8a4e5f24fc15f47c8d1b10d783cce7d0634840455b12d0ddfdfeb1a62a54c
Resume (Observed→InProgress)	Confirmed	faa7d80e3b9538dff557fde2d43fac7f44d1e1674c690cfadab61b7b0cb2c261
Certify (InProgress→Completed)	Confirmed	74058d465ec1bf0517f214f57b6eea27c07d0d85fd5a7065611e90bc48d40e98

Stage 9 — Terminaciones

Transition	Database status	TXID
Mint (declaration)	Confirmed	c06642bba288567c32f141bd5da62ee82a52c12c23864473ba4120b6a7f1bd88
Evidence anchor	Confirmed	ad7e1aefb94eb8a1b2867e33be8bf6d11fb79882bb9c8a0f2c008f912d9a1d0c
Automatic Pending→InProgress	Confirmed	35f7ce36629175fff331a418e1efadd917a0984fe86e02ad71d35b2c0da2bdbd
Observe (InProgress→Observed)	Confirmed	477a5c8bb0b2958ab8ace93f497fa2d8375b08b5d4fddfb79889a289f7dcca4
Resume (Observed→InProgress)	Confirmed	96b9f7ab479256e13dbc9bdd35bc553903205e7eee900c7b9717395a6835b803
Certify (InProgress→Completed)	Confirmed	5ed43e6c8240e65a654d21f85dee44c622ced1fb1bd287f76290893dd171b1de

Stage 10 — Final de obra y subdivisión

Transition	Database status	TXID
Mint (declaration)	Confirmed	93f98d71bede9119dd49958f9ec002b879e542dafa53558f4a3605215911ff4f
Evidence anchor	Confirmed	c34314d4276c030a43919d9d0f13b82edb3f36cb2e90cc19dfdde6e253f4a3ed
Automatic Pending→InProgress	Confirmed	c6d7abb1f40c74f8e5a185e152c37bf27ea1f3c2370cdaba0a6614b484eca841
Observe (InProgress→Observed)	Confirmed	4875496d4d31421226fca6216e41225a8c6f19fb29c5b633bdb4732e2e76dbe0
Resume (Observed→InProgress)	Confirmed	9ba0f810eb545ddf0060c9d092cdba0cbc563b53853c6509f9ec75f85b6da3ca
Certify (InProgress→Completed)	Confirmed	937b31b71561537cce17ea8a50c85e85c5f69c8af679e8bff6763eb22cf81ab3

Torre Volumen 3

Stage 1 — Adquisición del terreno

Transition	Database status	TXID
Mint (declaration)	Confirmed	478b903123df671aa422103d75b89a5ef8fce216fb2888518f471d2451b1fee8
Evidence anchor	Confirmed	439b10c4d6cde16501854011a7326c59b5315157c2b04708c1e23196398ede2e
Automatic Pending→InProgress	Confirmed	2d8e33cb04a1e05b7f552e0c6e143adc149c7f55daad9923dec1417605e6c830
Observe (InProgress→Observed)	Confirmed	9c54abe7db22e85d1bcbcc0ca525142e41b104f4c1be8f711771377d4e0bc358
Resume (Observed→InProgress)	Confirmed	17881ea34c88ada58eef0bb6d77533243a4da08fed8ecfde0d16894509f486a2
Certify (InProgress→Completed)	Confirmed	1d0bc8acd3918d6f35801a183e91cfe74cf789fe12835357cb706d02765c0029

Stage 2 — Proyecto ejecutivo

Transition	Database status	TXID
Mint (declaration)	Confirmed	3144dac1045e54597f3df797b097b87a660c26477ba0595b91a68bc2be80d9ab
Evidence anchor	Confirmed	f25230d359ca45f5aa0045412577cfc533f38bddc875ddc9ead7a0d131dc1ff6
Automatic Pending→InProgress	Confirmed	2195d5cc0e277172f5e76cc2b2e4c2453f8913481d1192eb156e8ce2cd5727c1
Observe (InProgress→Observed)	Confirmed	9f29b951d7eb9d2b03328b341dcf970a5b51bf3632324007e10c032dfd4850be
Resume (Observed→InProgress)	Confirmed	4a160f105af3eb4df18d96523240cde2c28ed14bf33c0ab84983e698c5e4e402
Certify (InProgress→Completed)	Confirmed	02ccd48a5eceb2e511ab45adf6aa1d261f04791cb389a9a66332f9c760a80bc9

Stage 3 — Movimiento de suelos y excavación

Transition	Database status	TXID
Mint (declaration)	Confirmed	bff850b52b5c2884952da8731c9599b39eb7602a60171e42c79695ed00a3391b3
Evidence anchor	Confirmed	835b6b83fb1edf009fe97f9d7197b0f111acc77da3acccd560cdc01a72abc49f
Automatic Pending→InProgress	Confirmed	4b07aad729cd5a8861fdbcf00da201669ca333cb90d895d43d5a70e62b87c347

Transition	Database status	TXID
Observe (InProgress→Observed)	Confirmed	6657dcfdcb1614ee66f7c9c525c1f5f285e242076c2f222fea13916ca92bad2f
Resume (Observed→InProgress)	Confirmed	7948229bd3215f47760142496f48518a1399fd40aff113acee1ece60ec78c452
Certify (InProgress→Completed)	Confirmed	13de262947dc69cf7e8e63c02b373dbfcdb8436a1a0d3855935d2f0a64b499ba

Stage 4 — Cimentación

Transition	Database status	TXID
Mint (declaration)	Confirmed	9b1bf8420d87c8ca697574ff3dfa2dd107c5ad5de54d4e6a07aa802f8a11abd1
Evidence anchor	Confirmed	ac7625e149a06742f57d7cc5a8ab215647d96b880ce35e0ef39fb08662d12248
Automatic Pending→InProgress	Confirmed	39d31941d695af5fc4cb5dd9e14d9331d6ab4a00e674d06eeb6ea3d69f5817cf
Observe (InProgress→Observed)	Confirmed	eab045b18ade0bda509b1bf52c63e4495feec0aa00757c1708136786b455d8ce
Resume (Observed→InProgress)	Confirmed	a91c57f134f0431df4d3065684a3c8d1ff7ae374d7fef6295bee6f4295bc9d4d
Certify (InProgress→Completed)	Confirmed	f2e687527c8624bf006a5fd3708e98d4fa9c301d7ad2c943cb90d7907be93dfb

Stage 5 — Estructura planta baja

Transition	Database status	TXID
Mint (declaration)	Confirmed	14a043d42ae6b6d0167edd3ffc1622dd057bd5aea14fe7cfc41ca5e6f8237e62
Evidence anchor	Confirmed	237284c136ae30880d9dbad12ac912271f4d448356e9043ef5fb53a2dccc6769
Automatic Pending→InProgress	Confirmed	b21dfeab948f0034759dba2d56b1f931ed63a6658535fde0128abb736774811e
Observe (InProgress→Observed)	Confirmed	1f4d989bdceb2d8108ea5c52268badcce18d9c043856ab093ff965f5b76636e8
Resume (Observed→InProgress)	Confirmed	aab113b3703f4022cab4c952ac1bb0de79f4e91dbebdeaef1e1fbbddcd10b9b
Certify (InProgress→Completed)	Confirmed	81bffe18c3f0c4463f9317a72d291df2980b661443b7bd46332673639d8baab5

Stage 6 — Estructura niveles superiores

Transition	Database status	TXID
Mint (declaration)	Confirmed	0c12c54224a65068a2d61fb8512f20690f2ba1973bee572a8ade476624c6a64f
Evidence anchor	Confirmed	6d1892ea8a7029e9309df5d39ccc340204c99189cc8939eafc1fecf5de94aedc
Automatic Pending→InProgress	Confirmed	60752955067bc36bdf48d76b501ef3712037e91659e28c8f7e7f1e480c70c52a
Observe (InProgress→Observed)	Confirmed	b47e4a9a55249dac7be2826793b89df0c3fd7e90419d4bc0dd7660c2e1987baf
Resume (Observed→InProgress)	Confirmed	6975287766f9dec81075afb6563b8e5edb9f0e33087cac49c32fd99444a5de66
Certify (InProgress→Completed)	Confirmed	c69626d201aa774d2f8235366f6efc9ebae5d975c333309f528ced956d1b8dcc

Stage 7 — Cerramientos y mampostería

Transition	Database status	TXID
Mint (declaration)	Confirmed	4d170440d8b8b916dc339bca9f0f9069d9cf34bf0df17ddc67a3d69b5e481498
Evidence anchor	Confirmed	dfb63807a7a2a15c112ff2284df7be28a25703af692cb4f0b394f4ed650b1c2d
Automatic Pending→InProgress	Confirmed	eb69ceb31f2152cb95dad6a87d5281acb30a4cf919ca6b644a81ef7c8a656149
Observe (InProgress→Observed)	Confirmed	7d2318bf3b5aaee4afd6733734242ac1557225a7b2d9a142bf688bb7c822329d
Resume (Observed→InProgress)	Confirmed	7a9b534233941030028d500af7d4a45e0d9b14832302fb9261705dc550531ed9
Certify (InProgress→Completed)	Confirmed	207f2b4b8ebc56ecae155ed9c1d1bea2a713f709822fdb30a01bed13743fa885

Stage 8 — Instalaciones

Transition	Database status	TXID
Mint (declaration)	Confirmed	d745e249f63ee6541e8926299ff5b90b4478996f27fb58614cabbb61a28b21a7
Evidence anchor	Confirmed	36ebc69e2199fbe8da36d5d43499aebc3a84404e78da4dffff1e6371ed588e914
Automatic Pending→InProgress	Confirmed	2d0e210b6c3781c1024e118882571d3986a27cb06e180f05170013325215eb9b
Observe (InProgress→Observed)	Confirmed	1a5399c3d5261967e7c18c8230374201f298b294b04a5e177236dd41ae82e6d2
Resume (Observed→InProgress)	Confirmed	1efb88123d2833aaf00838299dcf30bcb8dbe46f578ff4711b5a7d4b746b3734

Transition	Database status	TXID
Certify (InProgress→Completed)	Confirmed	fdf533c69cc1a0c2773f0e3938464c09f435e13dd4555d5bee8d64aed0e0001e

Stage 9 — Terminaciones

Transition	Database status	TXID
Mint (declaration)	Confirmed	0dae3d845591855b666ffd1997124c4ca640dae617481af879a2acd9574b58fd
Evidence anchor	Confirmed	3c08bf9dc899eb7c21b5d80f4f6c08d364a5a32b5ad5d6722ea6a30754a0d1eb
Automatic Pending→InProgress	Confirmed	385c2e0192531aa42d5a93fcf89d0c810fab71e3e60342222c8bf9155d2da0a8
Observe (InProgress→Observed)	Confirmed	5f918bc177f99aadca44cde11f1abd2c5d25853ba0acab304051271acc1d7ac5c
Resume (Observed→InProgress)	Confirmed	44486b07ae68be5fdb7132541556bf84d45da5b8b1640dc dab7d3649304440c5
Certify (InProgress→Completed)	Confirmed	792c435dd3827a92c1eaebba7330eb97ccdba9129178968b378d849ae04878ca

Stage 10 — Final de obra y subdivisión

Transition	Database status	TXID
Mint (declaration)	Confirmed	f0bdc168eedc9f1c473c82964ccd79e65e9f4b326ced5eb01940afa a65dab563
Evidence anchor	Confirmed	f36f1e5778943825ef55e2366097ad7e7c79fef40e48b65f913df9a c90777e4b
Automatic Pending→InProgress	Confirmed	692e380288714ac0c2d97a1e8e266e075c793a0e60feb4e48d2fcba 498c2b43f
Observe (InProgress→Observed)	Confirmed	b4b18c182592fae2770b188fa22e107910133c656054c00b55aca17 346d41a21
Resume (Observed→InProgress)	Confirmed	03007f19893539bafd50eb8ea4f9f15493192d5adb4f3a68650245c d1575f660
Certify (InProgress→Completed)	Confirmed	6a9d502c789bb291f1fd670a748777fa70a2d16645d77b9e38838ce 6bb8e66c2